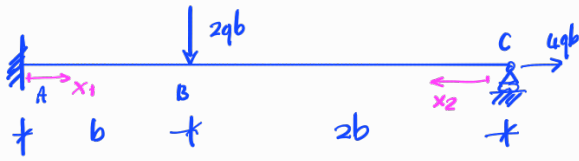


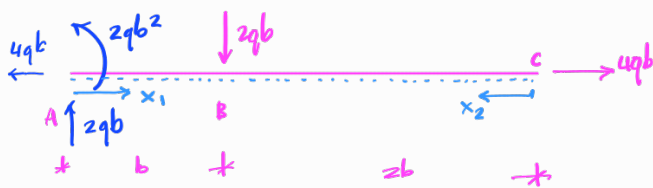
Struttura iperstatica



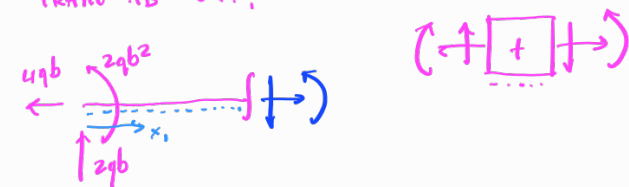
Struttura iperstatica



$$\begin{cases} R_x = 0 \rightarrow H_A + 4qb = 0 & \boxed{H_A = -4qb} \\ R_y = 0 \rightarrow V_A - 2qb = 0 & \boxed{V_A = 2qb} \\ M_A = 0 \rightarrow M_A - 2qb \cdot b = 0 & \boxed{M_A = 2qb^2} \end{cases}$$



TRATTO AB $0 < x_1 < b$



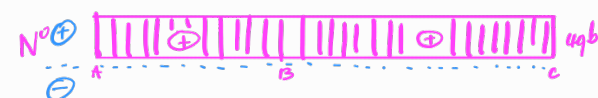
$$\begin{cases} N_{AB}^0 = 4qb \\ T_{AB}^0 = 2qb \\ M_{AB}^0 = -2qb^2 + 2qb x_1 \end{cases} \quad \begin{cases} x=0 & M_{AB}^0 = -2qb^2 \\ x=b & M_{AB}^0 = 0 \end{cases}$$

TRATTO CB $0 < x_2 < b$

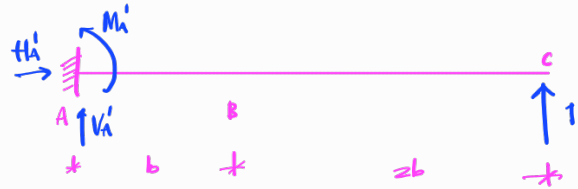


$$\begin{cases} N_{CB}^0 = 4qb \\ T_{CB}^0 = 0 \\ M_{CB}^0 = 0 \end{cases}$$

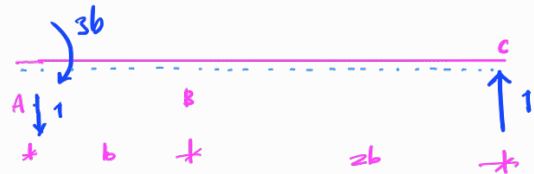
Diagrammi



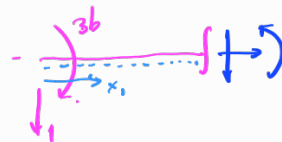
Struttura ausiliaria



$$\begin{cases} R_x = 0 \rightarrow \boxed{H_A' = 0} \\ R_y = 0 \rightarrow V_A' + 1 = 0 & \boxed{V_A' = -1} \\ M_A = 0 \rightarrow M_A' + 1 \cdot 3b = 0 & \boxed{M_A' = -3b} \end{cases}$$



TRATTO AB $0 < x_1 < b$



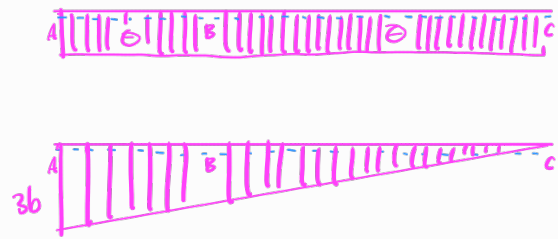
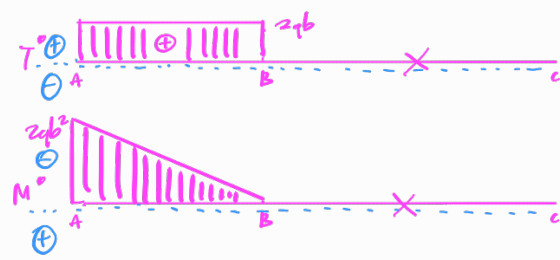
$$\begin{cases} N_{AB}' = 0 \\ T_{AB}' = -1 \\ M_{AB}' = 3b - 1 \cdot x_1 \end{cases} \quad \begin{cases} x=0 & M_{AB}' = 3b \\ x=b & M_{AB}' = 2b \end{cases}$$

TRATTO CB $0 < x_2 < b$



$$\begin{cases} N_{CB}' = 0 \\ T_{CB}' = -1 \\ M_{CB}' = 1 \cdot x_2 \end{cases} \quad \begin{cases} x=0 & M_{CB}' = 0 \\ x=b & M_{CB}' = 2b \end{cases}$$





$$\delta \mathcal{L}_e = \delta \mathcal{L}_i$$

$$\delta \mathcal{L}_e = 0$$

$$\delta \mathcal{L}_i = \int_0^L N'(x) \epsilon_x dx + \int_0^L T'(x) \chi_x dx + \int_0^L M'(x) \chi_x dx = 0$$

$$\chi = \frac{M^0(x) + X M'(x)}{EJ} \quad \rightarrow \quad \delta \mathcal{L}_i = \int_0^L M'(x) \frac{M^0(x) + X M'(x)}{EJ} dx = \int_0^L \frac{M'(x) M^0(x)}{EJ} dx + X \int_0^L \frac{M'(x)^2}{EJ} dx = 0$$

TRATTO	L	$M^0(x)$	$M'(x)$	$M'(x) M^0(x)$	$M'(x)^2$
AB	b	$-2qb^2 + 2qb^2x$	$3b - x$	$-6qb^3 + 2qb^2x + 6qb^2x - 2qb^2x^2$	$9b^2 - 6bx + x^2$
CB	2b	0	x	0	x^2

$$\begin{aligned} \mathcal{L}_i &= \int_0^b \frac{1}{EJ} (-6qb^3 + 2qb^2x + 6qb^2x - 2qb^2x^2) dx + X \int_0^b \frac{1}{EJ} (9b^2 - 6bx + x^2) dx + X \int_0^{2b} \frac{1}{EJ} (x^2) dx = 0 \\ &= \frac{1}{EJ} \left[-6qb^3x + 2qb^2 \frac{x^2}{2} + 6qb^2 \frac{x^2}{2} - 2qb^2 \frac{x^3}{3} \right]_0^b + \frac{1}{EJ} \left[X \left(9b^2x - 6b \frac{x^2}{2} + \frac{x^3}{3} \right)_0^b + X \left(\frac{x^3}{3} \right)_0^{2b} \right] = 0 \\ &= \frac{1}{EJ} \left(-6qb^4 + qb^4 + 3qb^4 - \frac{2}{3}qb^4 \right) + X \left(9b^3 - 3b^3 + \frac{b^3}{3} + \frac{8b^3}{3} \right) = 0 \end{aligned}$$

$$X = \frac{8}{27} qb$$

Struttura finale



$$\begin{cases} H_A = H_A^0 + X H_A^1 \rightarrow H_A = 4qb \\ V_A = V_A^0 + X V_A^1 \rightarrow V_A = \frac{46}{27} qb \\ M_A = M_A^0 + X M_A^1 \rightarrow M_A = \frac{10}{9} qb^2 \end{cases}$$

$$\begin{cases} N = N^0 + X N^1 \\ T = T^0 + X T^1 \\ M = M^0 + X M^1 \end{cases}$$

